Iniziato	martedì, 8 luglio 2025, 14:58
Stato	Completato
Terminato	martedì, 8 luglio 2025, 15:58
Tempo impiegato	1 ora

Domanda 1

Completo

Punteggio max.: 6,00

Construct a deterministic TM of the kind you prefer, which decides the following language:

$$\square = \{w \in \{0,1\}^* \mid w = (010)^n \text{ and } n \geq 0\}.$$

As an example 010010010 or the empty string are in □ while 01001 is not in □. Study the complexity of the TM you have defined.

```
ALPHABET={> (start), _ (blank), 0, 1}
STATES={q_init, q_read, q_0, q_01, q_010, q_rej, q_acc}

TRANSITION FUNCTION:
(q_init, >) --> (q_read, >, R)

(q_read, 0) --> (q_0, 0, R)
(q_read, 1) --> (q_rej, 1, S)
(q_read, _) --> (q_acc, _, S)   # empty string

(q_0, 0) --> (q_rej, 0, S)
(q_0, 1) --> (q_01, 1, R)
(q_0, _) --> (q_rej, _, S)

(q_01, 0) --> (q_010, 0, R)
(q_01, 1) --> (q_rej, 1, S)
(q_01, _) --> (q_rej, _, S)

(q_010, 0) --> (q_0, 0, R)
(q_010, 1) --> (q_rej, 1, S)
(q_010, _) --> (q_acc, _, S)

The complexity of my TM is linear as it scans the input once.
```

Domanda 2

Completo

Punteggio max.: 7,00

You are required to prove that the following problem □ □□□ is in **EXP**. To do that, you can give a TM or define some pseudocode. The language □ □□□ includes precisely those binary strings which are encodings of valid CNFs, namely of any CNF F such that $[\![F]\!]_\varrho = 1$ for every ϱ.

```
INPUT: F=[c_1, ..., c_n] where c_i is the list of at most h literals out of k literals in the form (name, polarity)
OUTPUT: True is for every assignment the CNF is sat

PSEUDOCODE:
for val_1 in {True,False}:
    ...
        for val_k in {True,False}:
            assignment={name_1:val_1, ..., name_k:val_k}
            for c in F:
                sat = False
                for i in range(0,len(c)):
                    lit = c[i][0]
                    pol = c[i][1]
                    real_value = assign[lit] if pol else not assign[lit]
                    sat = sat or real_value
                 if not sat:   # the whole formula is unsat, try a new assignment
                    break
              if sat:
                 return True
return False

The pseudocode works in O(2^k * |F| * |c_i|) = O(2^k*n*h) so exponentially bounded to the input, so the problem is in EXP
```

Commento: Overall the solution is correct, but you return True if only one assignment makes the formula true, and False otherwise. The definition of TAUT is in fact the dual one!

Domanda 3

Completo

Punteggio max.: 7,00

Consider the following language:

$$\mathtt{VARSAT} = \{s \in \{0,1\}^* \mid s \text{ is the encoding of a triple } (F, G, H) \text{ of CNFs at least one of which is satisfiable}\}.$$

What is the complexity of $\mathtt{VARSAT}$?

INPUT: T=[F,G,H]
OUTPUT: True/False if (F or G or H) is sat

CERTIFICATE: an assignment of variables for each CNF, so A=[f, g, h] where each of them is an assignment for the variables of its formula

VERIFIER:

```
for i in range(0,3):
    F = T[i]
    a = A[i]
    for c in F:
        sat = False
        for j in range(0,len(c)):
           lit = c[j][0]
           pol = c[j][1]
           real_value = a[lit] if pol else not a[lit]
           sat = sat or real_value
         if not sat:   # the whole formula is unsat, try a new assignment
           break
     if sat:
        return True
return False
```

1) The certificate is a list of dictionaries in the form {literal_name:literal_polarity} so polynomially bounded to the number of literals, so to the input.
2) The verifier works in O(3*|F_i|*|c_j|) so polynomially bounded to the input.

Moreover we can prove the reduction SAT <_p VARSAT, since we can find a polytime reduction f so that F is sat <==> f(F) is varsat where f(F) = (a /\ -a) \/ F \/ (b /\ -b) so we add 2 trivial unsat 1CNF as G and H and then the 2 problems are equal, proving the bidirectional arrow.

◄ ICC - 08072025 - QUESTIONS

Vai a...